

Muhammad Umer Khan Niazi

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EDUCATION

University of Engineering and Technology

BS Artificial Intelligence

Lahore, Pakistan

Expected 2029

Forman Christian College

Intermediate Computer Science

Lahore, Pakistan

2022 – 2024

PROJECTS

News NLP Pipeline | *Python, SQLite, BERTopic, spaCy, Transformers, Streamlit*

2026 – Present

- Built an end-to-end NLP pipeline that discovers topics, measures sentiment, and extracts named entities from 15 years of Dawn News headlines without any manual labeling
- Scraped and processed 350K+ headlines; applied BERTopic for unsupervised topic discovery across historical news data, surfacing 20 distinct topic clusters
- Extracted 420K+ entity mentions and deployed an interactive Streamlit dashboard for exploring results

Himalayan Wildlife Classifier | *Python, PyTorch, ResNet18, Streamlit*

2026 – Present

- Built a PyTorch image classifier for Snow Leopard, Markhor, and Himalayan Brown Bear using transfer learning on a frozen ResNet18 backbone with a custom 4-class classification head
- Achieved 96.55% validation accuracy; added an internal Other class and 80% confidence threshold to reject out-of-distribution inputs and reduce false positives
- Deployed an interactive Streamlit interface for real-time inference with per-class confidence scores

wmus | *Python, windows-curses, pygame*

2026 – Present

- Built a keyboard-first terminal music player for Windows that recreates the cmus workflow natively, filling a gap left by cmus being unavailable on the platform
- Implemented fuzzy search with intelligent filtering, smart metadata caching for fast library startup, and fully configurable keybindings
- Packaged as a standalone Windows installer using PyInstaller and Inno Setup for distribution outside the Python ecosystem

TECHNICAL SKILLS

Languages: Python, C++, GDScript

Systems & Tools: Linux, Git, SQLite, Bash

Machine Learning & Data: PyTorch, Transformers, scikit-learn, Hugging Face, spaCy, BERTopic, NLTK, Pandas, NumPy

Development & Deployment: Godot, Unity, Streamlit, GitHub Pages